

AMENDMENT NO. 2 MAY 2020

TO

**IS 14164 : 2008 INDUSTRIAL APPLICATION AND
FINISHINGS OF THERMAL INSULATION
MATERIALS AT TEMPERATURES ABOVE
–80°C AND UP TO 750°C — CODE OF PRACTICE**

(First Revision)

(Page 16, clause 6.2.4) — Insert the following new clauses:

6.2.5 Finishing Cement

The finishing cements shall have a minimum thickness of 10 mm or as specified and shall be reinforced with wire netting of maximum 20 mm mesh and minimum 0.56 mm wire diameter or equivalent metal lath.

6.2.6 Bituminous Mastics

Minimum of 3 mm dry thickness, inclusive of the reinforcement, with suitable reinforcement where necessary.

6.2.7 Roofing Felt

Self finish, conforming to Type 3, Grade 1 of IS 1322.

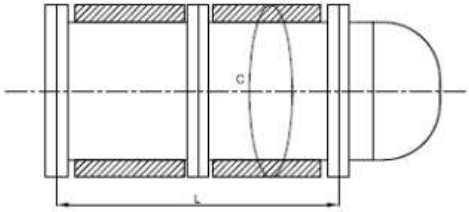
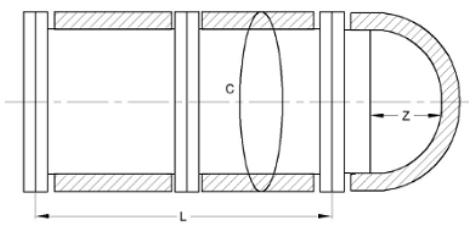
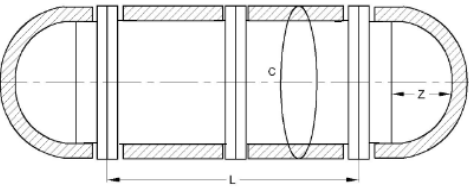
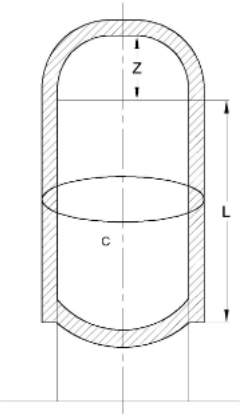
6.2.8 Other Finishes

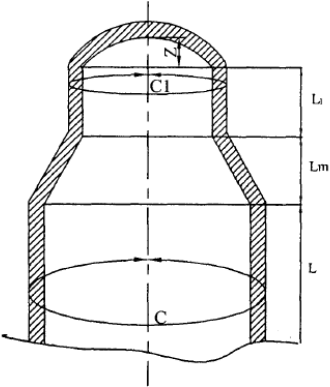
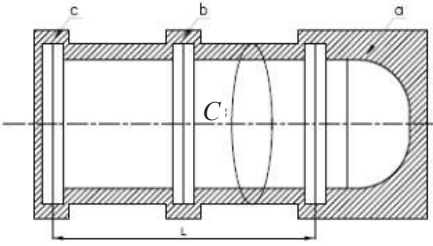
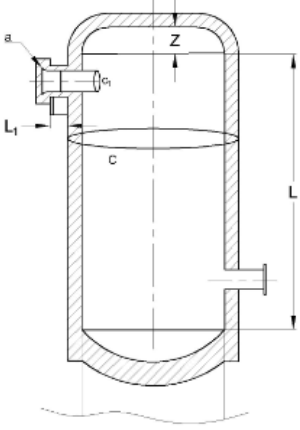
Where conditions of exposure are not very severe, low cost outer coverings such as jute canvas (weighing not less than 0.34 kg/m²), cotton canvas (weighing not less than 0.27 kg/m²) and scrim cloth (loose weave hessian weighing not less than 0.13 kg/m²) may also be used.

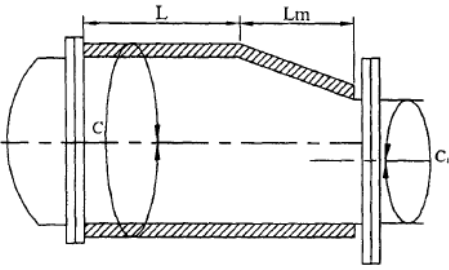
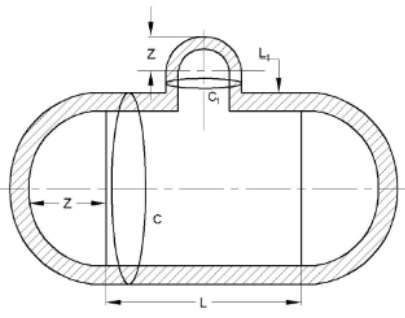
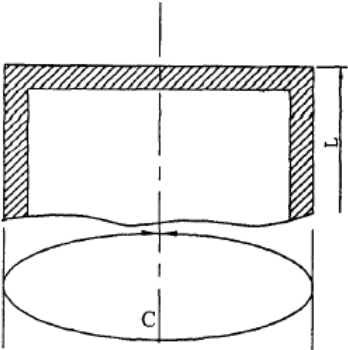
NOTE — 6.2.7 and 6.2.8 are generally associated with insulation for hot surfaces.

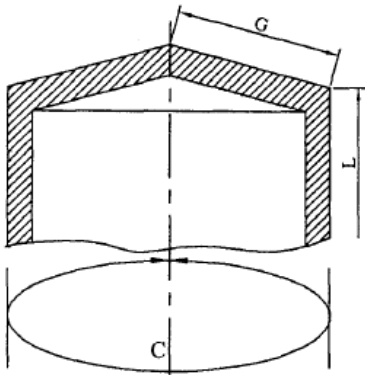
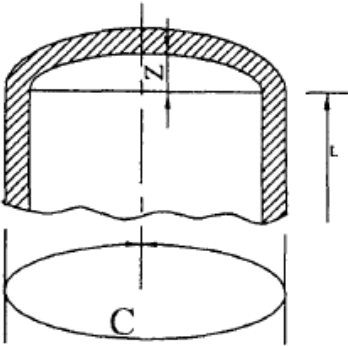
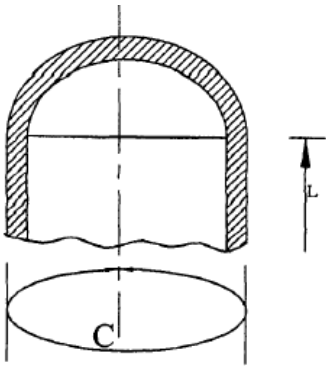
Price Group 2

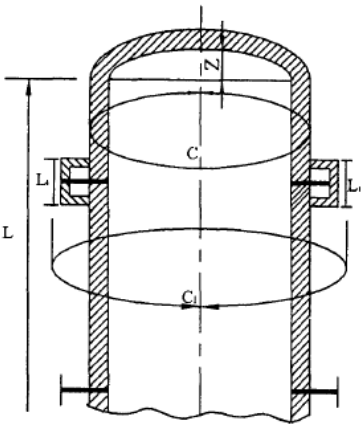
[Page 21-25, clause 9.2.1] — Substitute the following table for the existing:

	Typical Exemplifications	Relevant Surfaces
Fig. 13		CL
Fig. 14		$CL + \frac{\pi D^2}{4} Y$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$
Fig. 15		$CL + \frac{\pi D^2}{2} Y$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$
Fig. 16		$CL + \frac{\pi D^2}{2} Y$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$

	Typical Exemplifications	Relevant Surfaces
Fig. 17		$CL + \frac{C + C_1}{2} L_m Y$ $+ C_1 L_1 + \frac{\pi D_1^2}{4} Y_1$ $Y = 1.2$ and $Y_1 = 1.27$ where $Z < D/3$ and $Y_1 = 1.75$ where $Z \geq D/3$
Fig. 18		$C(L + X_a + X_b + X_c)$ where $X_a = 1$ $X_b = 0.5$ $X_c = 0.5$
Fig. 19		$CL + C_1(L_1 + X_a)$ $+ \frac{\pi D_1^2}{2} Y$ where $X_a = 0.5$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$

Typical Exemplifications		Relevant Surfaces
Fig. 20		$CL + \frac{C + C_1}{2} L_m Y$ where $Y = 1.2$
Fig. 21		$CL + \frac{\pi D^2}{2} Y + C_1 L_1 + \frac{\pi D^2}{4} Y$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$
Fig. 22		$CL + \frac{\pi D^2}{4}$

Typical Exemplifications		Relevant Surfaces
Fig. 23		For $G > 3m$: $CL + \pi G^2$ For $G < 3m$: $CL + \pi G^2 Y$ where $Y = 1.5$
Fig. 24		$CL + \frac{\pi D^2}{4} Y$ $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$
Fig. 25		$CL + \frac{\pi D^2}{4} Y$ where $Y = 1.27$

Typical Exemplifications		Relevant Surfaces
Fig. 26		$CL + \frac{\pi D^2}{4} Y$ $+ C_1 L_1 Y_1^4$ $Y_1 = 1.2$ and $Y = 1.27$ where $Z < D/3$ and $Y = 1.75$ where $Z \geq D/3$

[Page 26, clause 9.3.2.2, line 4] — Insert ‘or Table 3, as applicable’ after ‘according to Table 2’.

[Page 26, Table 2] — Insert the following new table after Table 2.

**Table 3 Conventional Equivalent Lengths for Other Parts,
for Other Finishes**
(Clause 9.3.2.2)

Sl No.	Insulated Fitting	Equivalent Length
(1)	(2)	(3)
i)	Valve/venturi/steam trap/strainers including flange and body (upto 300 mm size)	1.40 m
ii)	Valve/venturi/steam trap/strainers including flange and body (for sizes larger than 300 mm)	1.90 m
iii)	Pair of flanges including orifice plate and flanges	0.60 m

Table 3 (Concluded)

Sl No.	Insulated Fitting	Equivalent Length
(1)	(2)	(3)
iv)	Bends and elbows	Twice the actual length, as measured along the center line of the piping or ducting
v)	Reducer	Actual length of larger size (along the center line of piping)
vi)	Tees	$2(D_1 + D_2)$; where D_1 and D_2 are insulated diameters of the two pipelines forming the tee
vii)	Any other special fitting	To be specified and measured separately on number basis

NOTE — Measurements of all valves, flanges and other fittings shall be based on actual content and then shall be converted into equivalent lengths of connected piping, to arrive at the total equivalent lengths of piping or ducting of various diameters. Fittings that connect two or more different sizes of piping/ducting shall be counted as part of the larger size.

(Page 28, Fig. 28, Caption) — Substitute ‘NB $\geq$ 50 mm’ for ‘NB < 50 mm’.